project.org

STUDENT HANDOUT

GREEN LEVEL
MORE SUPPORT

The Science of Seeing

Grades 5–8 · Science

ADMIT ONE

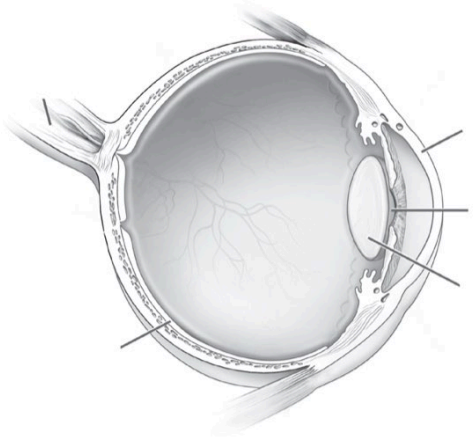
NAME _____ DATE _____ CLASS _____

How your eyes work. Your eyes collect light, but your **brain** creates the picture you see! Light travels through your eye and gets turned into electrical signals that your brain interprets.

01 PART 1 — Label the Eye

Use the **Word Bank** to label the parts of the eye diagram below.

Word Bank: Cornea · Pupil · Lens · Retina · Optic Nerve



1. _____
   (clear front part, bends light)
2. _____
   (black circle, controls light amount)
3. _____
   (focuses light onto back of eye)
4. _____
   (light-sensitive layer with rods and cones)
5. _____
   (carries signals to brain — location of your blind spot!)

02 PART 2 — The Blind Spot Experiment

Use the Blind Spot Test Card your teacher gave you. Follow these steps:

1. Close your LEFT eye **2.** Look at the + with your RIGHT eye **3.** Slowly move the card closer to your face **4.** The dot should disappear!

Why does this happen? Circle the correct answer:

A) The dot moves off the card B) Your optic nerve has no light receptors (blind spot)

C) You blinked

PART 3

03 Two Types of Illusions

Eye-based illusions: Trick the cells in your eye (like afterimages that tire out your cones)

Brain-based illusions: Trick how your brain interprets what it sees (like size illusions)

PART 4

04 Match the Illusion!

Draw a line from each illusion to where it happens:

Color Afterimage ●	● **EYE (retina)**
Size Illusion ●	● **BRAIN (visual cortex)**
Blind Spot ●	

MINI MUSEUM CONNECTION

The **stereoscope** in the Mini Museum tricks your brain! It shows each eye a slightly different picture. Your brain combines them into one 3D image.

Try the Perception Lab: minimuseumproject.org/exhibitions/seeing-is-deceiving/illusions

minimuseumproject.org/exhibitions/seeing-is-deceiving

PART 5

05 What I Learned

Fill in the blanks:

My eyes collect _____ , but my _____ creates the picture I see.

The blind spot is where the _____ _____ connects to my eye.

Optical illusions work by tricking either my _____ or my _____ .

SHARE WITH YOUR FAMILY

Try the blind spot experiment with your family! Can they find their blind spot too?

Fun fact to share: Your blind spot is about the size of 17 full moons in the sky — but you never notice it because your brain fills it in!